

AWS CLOUD ARCHITECTING	
DURATION	60 Hrs
INTENDED AUDIENCE	Students of AWS Academy institution
STUDENT PREREQUISITE	General IT Technical & Business Knowledge
DELIVERY METHOD	Lecture & Digital Training Model
LEARNING RESOURCES	Lecture Materials
	Online Multiple-Choice Knowledge Checks
	Lab Exercises
	Video Instructions/Demos
	Example Solutions
	Course Assessment
INTERNATIONAL CERTIFICATION	SAA-C03

S.No	Module Section
1	Welcome to AWS Academy Cloud Architecting
	Course Overview
	Café Business Case Introduction
	Roles in Cloud Computing
2	Introducing Cloud Architecting
	Cloud Architecting
	AWS Well-Architected Framework
	Best Practices for Building Solutions on AWS
	AWS Global Infrastructure
3	Securing Access
	Security Principles
	Authenticating and Securing Access
	Authorizing users
	Parts of an IAM Policy
	Activity: Examining IAM Policies
4	Guided Lab: Exploring AWS Identity and Access Management (IAM)
	Adding a Storage Layer with Amazon S3
	Defining Amazon S3
	Using Amazon S3
	Moving data to and from Amazon S3
	Demo: Amazon S3 Transfer Acceleration
	Storing content with Amazon S3
	Demo: Managing Lifecycles in Amazon S3
	Demo: Amazon S3 Versioning
	Designing with Amazon S3
	Activity: Designing with Amazon S3
	Challenge lab: Creating a Static Website for the Café
	Applying the AWS Well-Architected Framework principles to storage
	Adding a Compute Layer using Amazon EC2
	Adding compute with Amazon EC2
	Choosing an AMI to launch an EC2 instance
	Selecting an EC2 instance type
	Activity: Choosing Instance Types
	Adding storage to an Amazon EC2 instance

5	Guided lab: Introducing Amazon Elastic File System (Amazon EFS)
	Other EC2 configuration considerations
	Demo: Configuring an EC2 Instance with User Data
	Amazon EC2 pricing options
	Demo: Reviewing the Spot Instance History Page
	Challenge lab: Creating a Dynamic Website for the Café
	Applying the AWS Well-Architected Framework principles to compute
6	Adding a Database Layer
	Database layer considerations
	Amazon RDS
	Guided lab: Creating an Amazon RDS Database
	Amazon RDS Proxy connection management
	Demo: Amazon RDS Automated Backup and Read Replicas
	Amazon DynamoDB
	Purpose-built databases
	Migrating data into AWS databases
	Challenge lab: Migrating a Database to Amazon RDS
	Applying AWS Well-Architected Framework principles to the database layer
7	Creating a Networking Environment
	Introducing Amazon VPC
	Activity: Choose the Right Type of Subnet
	Securing network resources
	Demo: Creating an Amazon VPC in the AWS Management Console
	Connecting to managed AWS services
	Monitoring your network
	Applying AWS Well-Architected Framework principles to a network
	Guided lab: Creating a Virtual Private Cloud
	Challenge lab: Creating a VPC Networking Environment for the Café
8	Connecting Networks
	Scaling your VPC network with AWS Transit Gateway
	Activity: Configure AWS Transit Gateway Routes
	Connecting VPCs in AWS with VPC peering
	Guided lab: Creating a VPC Peering Connection
	Connecting to your remote network with AWS Site-to-Site VPN
	Connecting to your remote network with AWS Direct Connect
	Applying AWS Well-Architected Framework principles to network connectivity
9	Securing User, Application and Data Access
	Managing permissions
	Federating Users
	Guided Lab: Securing Applications by using Amazon Cognito
	Managing Access to Multiple Accounts
	Encrypting Data at rest
	Guided Lab: Encrypting Data at rest by using AWS Encryption Options
	AWS Security Services for Securing User, application, and Data Access
	Implementing Monitoring , Elasticity and High Availability
	Monitoring your resources
	Scaling your Compute resources
	Demo: Creating Scaling Policies for Amazon EC2 Auto Scaling

10	Scaling your Databases
	Using Load balancers to create highly available environments
	Demo: Creating a highly available application
	Using Route 53 to create highly available environments
	Using Route 53 to create highly available environments
	Demo: Amazon Route 53: Failover Routing
	Demo: Amazon Route 53: Geolocation Routing
	Guided lab: Creating a Highly Available Environment
	Challenge lab: Creating a Scalable and Highly Available Environment for the Café
	Applying AWS Well-Architected Framework principles to highly available systems
11	Automating Your Architecture
	Reasons to automate
	Using infrastructure as code
	Customizing with CloudFormation
	Demo: Analyzing an AWS CloudFormation Template
	Demo: AWS CloudFormation Resources
	Demo: Reviewing an AWS CloudFormation Template
	Demo: Using the AWS CloudFormation Console
	Using AWS Quick Starts
	Customizing with CodeWhisperer
	Guided lab: Automating Infrastructure with AWS CloudFormation
	Applying AWS Well-Architected Framework principles to automation
	Challenge lab: Automating Infrastructure Deployment
12	Caching Content
	Overview of caching
	Caching using CloudFront
	Guided lab: Streaming Dynamic Content Using Amazon CloudFront
	Caching using ElastiCache
13	Applying the AWS Well-Architected Framework principles to caching
	Building Decoupled Architectures
	Decoupling your architecture
	Decoupling applications with Amazon SQS
	Guided Lab: Building Decoupled Applications by using Amazon SQS
14	Decoupling applications with Amazon SNS
	Decoupling a hybrid application with Amazon MQ
	Building Serverless Architectures and Microservices
	Thinking Serverless
	Architecting Serverless Microservices
	Building Serless Architectures with AWS Lambda
	Demo: Using AWS Lambda with Amazon S3
	Guided lab: Implementing a Serverless Architecture on AWS
	Building microservice applications with AWS container services
	Demo: Running a Container
	Orchestrating microservices with AWS Step Functions
	Extending serverless architectures with Amazon API Gateway
	Activity: Decomposing a Monolithic Application with AWS API Gateway
	(Optional) Guided lab: Breaking a Monolithic Node.js Application into Microservices
	Applying AWS Well-Architected Framework principles to microservices and serverless

	Challenge lab: Implementing a Serverless Architecture for the Café
15	Data Engineering Patterns
	Data Characteristics
	Data Pipelines
	AWS tools to ingest data
	Processing batch data
	Processing real-time data
	Storage in the data pipeline
	Activity: Choosing Data Storage for a bank Application
	Parallel processing in the data pipeline
	Analysis and Visualization
	Activity:Data Pipeline Architecture
	Applying the AWS Well-Architected Framework Principles to data pipelines
16	Planning for Disaster
	Disaster planning strategies
	AWS disaster recovery planning
	Disaster recovery patterns
	Guided lab: Configuring Hybrid Storage and Migrating Data with AWS Storage Gateway S3
17	Applying AWS Well-Architected Framework principles to disaster planning
	Bridging to Certification
	Solutions Architect – Associate certification domains
	Solutions Architect – Associate certification exam resources
	Additional resources

